

Part 1

$$\begin{array}{lll}
 v_D = -\frac{eE\tau}{m} & U(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r} & \lambda = \frac{h}{p} = \frac{h}{mv} \\
 J = -N_e e v_D = \frac{N_e e^2 \tau}{m} E & \mathbf{F} = q\left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right) & \Delta p \Delta z \geq \frac{\hbar}{2} \\
 J = I/A & F_{\text{cent}} = \frac{mv^2}{r} & v_p = \frac{\omega}{k} \\
 \sigma = \frac{N_e e^2 \tau}{m} = N_e e \mu & \left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} = \mu \mathbf{J} + \mu \epsilon \frac{\partial \mathbf{E}}{\partial t} \end{array} \right. & \begin{array}{l} v_g = \frac{\partial \omega}{\partial k} \\ -\frac{\hbar^2}{2m} \nabla^2 \Psi + V \Psi = i \hbar \frac{\partial \Psi}{\partial t} \\ -\frac{\hbar^2}{2m} \nabla^2 \psi + V \psi = E \psi \end{array} \\
 \mu = \frac{e\tau}{m} = \frac{v_D}{E} & & \left[-\frac{\hbar^2}{2m_e} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} \right] \psi(\mathbf{r}) = E \psi(\mathbf{r}) \\
 R_H = \frac{E_H}{J_x B} = \frac{1}{N_e e} & & a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m e^2} \\
 c = \lambda v & & E_n = -\frac{Z^2 m e^4}{8 \epsilon_0^2 \hbar^2 n^2} \\
 & & E_n = -13.6 \text{ eV} \cdot Z^2 / n^2
 \end{array}$$

Part 2

$$\begin{array}{lll}
 \sigma = \frac{N_e e^2 \tau}{m^*} = N_e e \mu & u_k(x+a) = u_k(x) & N_v = 2 \left(\frac{2\pi m_h^* k_B T}{\hbar^2} \right)^{3/2} \\
 \mu = \frac{e\tau}{m^*} & \cos(ka) = P \frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a) & N_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right) \\
 E_{3D} = \frac{\hbar^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2) & P = \frac{ma}{\hbar^2} V_0 w, \quad \alpha = \sqrt{\frac{2mE}{\hbar^2}} & N_e \cdot N_h = N_i^2 \\
 Z(E) = \frac{dN}{dE} = \frac{4\pi L^3 (2m)^{3/2}}{\hbar^3} \sqrt{E} & v_g = \frac{1}{\hbar} \frac{\partial E}{\partial k} & N_e + N_A^- = N_h + N_D^+ \\
 f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)} & m^* = \hbar^2 \left(\frac{\partial^2 E}{\partial k^2} \right)^{-1} & E_F^{(\text{intr})} = \frac{E_g}{2} + \frac{3}{4} k_B T \ln\left(\frac{m_h^*}{m_e^*}\right) \\
 f(E) \approx \exp\left(-\frac{E - E_F}{k_B T}\right) & N_e = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) & E_F = E_c - k_B T \ln\left(\frac{N_c}{N_e}\right) \\
 \psi_k(x) = u_k(x) e^{ikx} & N_h = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right) & E_F = E_v + k_B T \ln\left(\frac{N_v}{N_h}\right) \\
 & N_c = 2 \left(\frac{2\pi m_e^* k_B T}{\hbar^2} \right)^{3/2} & E = \hbar \omega = hc/\lambda
 \end{array}$$

Part 3

$$\begin{array}{lll}
 Z(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} \sqrt{E - E_c} & V(x) = \begin{cases} \frac{eN_A}{2\epsilon} (x + x_p)^2 & -x_p < x < 0 \\ V_{bi} - \frac{eN_D}{2\epsilon} (x - x_n)^2 & 0 < x < x_n \end{cases} & I = I_0 \left[\exp\left(\frac{eU_1}{k_B T}\right) - 1 \right] \\
 N_A x_p = N_D x_n & V_{bi} = \frac{e}{2\epsilon} (N_A x_p^2 + N_D x_n^2) & I(d) = I_0 e^{-\alpha d} \\
 E(x) = \begin{cases} -\frac{eN_A}{\epsilon} (x + x_p) & -x_p < x < 0 \\ \frac{eN_D}{\epsilon} (x - x_n) & 0 < x < x_n \end{cases} & \frac{n_n}{n_p} = \frac{N_A N_D}{n_i^2} = \exp\left(\frac{eV_{bi}}{k_B T}\right) & \varepsilon(\omega) = \varepsilon'(\omega) - i \varepsilon''(\omega) \\
 & W = \sqrt{\frac{2\epsilon V_{bi}}{e} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)} & M = \chi_m H \\
 & & B = \mu_0 (H + M) = \mu_0 (1 + \chi_m) H \\
 & & \chi_m = \frac{C}{T - \theta}, \quad T > T_C
 \end{array}$$